Research Trend Analysis of NeurIPS Papers (2023–2025): A Review-Guided Topic Mining Study

Tran Hung Dat
VinUniversity
22dat.th@vinuni.edu.vn

Tran Ho Chi Thanh
VinUniversity
24thanh.thc@vinuni.edu.vn

Nguyen Pham Tuan Anh
VinUniversity
24anh.npt@vinuni.edu.vn

Our GitHub Repository

Abstract

We analyze how research topics in 12,539 accepted NeurIPS papers (2023–2025) evolved and intersected with peer-review signals. Because public rejected submissions are incomplete on OpenReview, we avoid unidentifiable topic-level acceptance rates, focusing instead on accepted-corpus quantities like yearly topic share and year-normalized reviewer ratings. Using an outlier-reduced BERTopic pipeline with MiniLM embeddings, we extract 56 topics that significantly outperform an LDA baseline in coherence (c_v: 0.733 vs. 0.526). Our main empirical finding reveals a notable review-trend misalignment: topic growth in accepted-paper share is negatively associated with year-normalized reviewer ratings (Pearson $r = -0.384$; Spearman $\rho = -0.471$). This negative association remains robust across controlled regressions, bootstrap confidence intervals, and adjacent-year transition analyses.

1 Introduction

Conference papers provide a useful record of how research interests evolve over time. By examining the topics represented in published papers, we can identify emerging areas, declining themes, and shifts in community attention. NeurIPS is particularly well suited for this type of analysis because it spans a broad range of machine-learning research and provides access to detailed review information through OpenReview. While simple keyword-frequency analyses can reveal commonly used terms, they often fail to capture coherent research themes or explain how topic popularity relates to the peer-review process. To address these limitations, we combine topic modeling with review-based analysis.

An important challenge in this project is ensuring that the quantities being analyzed can be reliably measured from the available data. In our milestone report, we referred to "topic acceptance rates." However, rejected NeurIPS submissions are not fully visible on OpenReview. As noted by Paper Copilot, publicly available rejected papers represent only the subset of authors who choose to release their submissions, whereas overall acceptance statistics are reported separately (1). Because the full set of rejected papers is unavailable, topic-level acceptance rates cannot be estimated accurately.

We therefore reformulate our research question: *among accepted NeurIPS papers, which topics are growing or declining over time, and how are these trends related to reviewer ratings and decision outcomes?* This revised focus allows us to study research trends using quantities that are directly

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observable from the data. Rather than relying solely on the difference between 2023 and 2025, we also examine year-to-year changes to capture shorter-term shifts in research activity.

Our contributions are fourfold. First, we develop a data-processing pipeline for collecting and cleaning NeurIPS metadata from OpenReview. Second, we construct and validate a BERTopic-based topic-modeling framework, including embedding comparisons, hyperparameter tuning, outlier reduction, benchmarking against LDA, and evaluation using coherence, diversity, and stability metrics. Third, we investigate the relationship between topic trends and peer-review signals through correlation analysis, regression models with robust standard errors, bootstrap confidence intervals, and transition-level analyses. Finally, we perform representative-paper validation to assess the interpretability and semantic consistency of the discovered topics.

2 Related Work and Positioning

Conference trend analysis. Conference-level analyses often track paper counts, keywords, or author-provided metadata. Such approaches are useful but can conflate true shifts in research attention with conference growth. We address this by using normalized yearly accepted-paper shares rather than raw counts.

Topic modeling and scientific embeddings. LDA is a classical topic model based on word co-occurrence and latent mixtures (6). BERTopic combines neural embeddings, UMAP, HDBSCAN, and c-TF-IDF to produce semantically coherent topics (3; 7; 8). Because our documents are scientific papers, we compare MiniLM sentence embeddings with SPECTER2-style scientific-document representations (5). We evaluate c-NPMI, c_v coherence, diversity, and assignment stability rather than relying on one visually plausible model.

Peer-review analysis. Review metadata are often studied separately from topic mining. Our project links the two: topic movement is measured from the accepted corpus, while reviewer ratings and decision tiers provide review-grounded signals. The analysis is descriptive rather than causal; it asks whether topics growing in accepted-paper share receive higher or lower review scores.

3 Dataset and Validity Correction

The crawl contains 13,171 public OpenReview records across NeurIPS 2023–2025. The accepted-paper analysis set contains 12,539 papers: 3,218 in 2023, 4,035 in 2024, and 5,286 in 2025. The public rejected subset contains 632 records; it is retained only for transparency, not for acceptance-rate estimation.

Table 1: Corrected dataset summary. Public rejected records are not used as an acceptance-rate denominator.

Year	Public records	Accepted	Public rejected
2023	3,395	3,218	177
2024	4,236	4,035	201
2025	5,540	5,286	254
Total	13,171	12,539	632

For topic c, the true acceptance rate would be

$$\alpha_c = \frac{A_c}{A_c + R_c^{public} + R_c^{hidden}},$$

where A_c is the accepted count, R_c^{public} is the visible rejected count, and R_c^{hidden} is the hidden rejected count. Since R_c^{hidden} is unobserved and may vary by topic, the visible ratio $A_c/(A_c + R_c^{public})$ is an upward-biased quantity, not a topic acceptance rate. The final report therefore removes all topic-level acceptance-rate statements and uses Paper Copilot only for global selectivity context.

4 Method

Preprocessing and embeddings. Each document is the concatenation of title and abstract. We remove URLs, LaTeX artifacts, citation markers, repeated whitespace, and generic noise tokens, then lowercase and lemmatize text. We compare all-MiniLM-L6-v2 for efficient semantic embeddings and SPECTER2 for scientific-document representations (5). Embeddings are cached so transformer inference is separated from clustering grid search.

Topic modeling. We use BERTopic, which combines document embeddings, UMAP dimensionality reduction, HDBSCAN density clustering, and c-TF-IDF topic representations (3; 7; 8). The grid varies embedding family, seed, UMAP neighbors/components, HDBSCAN minimum cluster size, and HDBSCAN minimum samples. HDBSCAN naturally assigns low-density documents to outlier topic -1; after choosing the best configuration, we apply BERTopic's embedding-based outlier reduction (4). We compare with an LDA TF-IDF baseline with $K = 30$ topics (6).

Trend and review metrics. Let $n_{c,y}$ be accepted papers in topic c and year y, and N_y be accepted papers in year y. We compute normalized accepted-paper share

$$\hat{s}_{c,y} = \frac{n_{c,y}}{N_y}.$$

Because the study covers only three annual observations, we treat endpoint change as net movement rather than long-term forecast:

$$\Delta_c = \hat{s}_{c,2025} - \hat{s}_{c,2023}.$$

To capture year-specific shocks, we decompose movement into

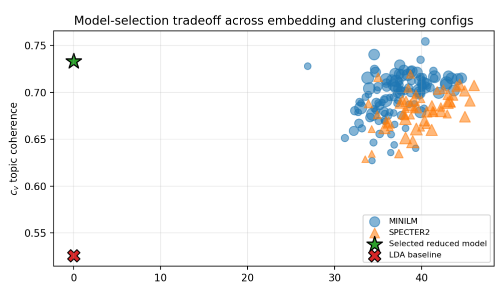
$$g_{c,23\to24} = \hat{s}_{c,2024} - \hat{s}_{c,2023}, \qquad g_{c,24\to25} = \hat{s}_{c,2025} - \hat{s}_{c,2024}.$$

Reviewer ratings are standardized by year, $z_i = (r_i - \mu_{y_i})/\sigma_{y_i}$, and prestige concentration is

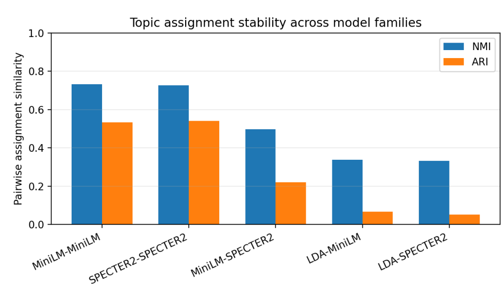
$$\pi_c = \frac{\#(\text{oral or spotlight papers in } c)}{\#(\text{accepted papers in } c)}.$$

5 Experiments and Model Selection

We ran the full pipeline on an NVIDIA RTX A5000 server. The grid produced 165 evaluated artifacts, including BERTopic configurations, the selected outlier-reduced model, and the LDA baseline. Invalid configurations with too few topic-level documents for c-TF-IDF vectorization were discarded. The selected pre-reduction model is MiniLM with seed 42, UMAP $n\_neighbors = 15$, UMAP $n\_components = 10$, HDBSCAN $min\_cluster\_size = 30$, and HDBSCAN $min\_samples = 10$. Outlier reduction assigns all accepted papers to 56 topics.



(a) Model-selection tradeoff.



(b) Assignment stability.

Figure 1: Model diagnostics. The selected outlier-reduced BERTopic model removes outliers while retaining high coherence. Same-family BERTopic runs are more stable than cross-embedding or LDA comparisons.

Pairwise stability shows that same-family BERTopic runs are much more similar than cross-embedding runs or LDA comparisons. MiniLM–MiniLM pairs have mean NMI 0.733 and mean ARI 0.533; SPECTER2–SPECTER2 pairs have mean NMI 0.727 and mean ARI 0.542; MiniLM–SPECTER2 pairs drop to mean NMI 0.497. Thus, embedding choice changes topic boundaries, but each embedding family is internally stable.

Table 2: Model selection and baseline comparison. Outlier reduction improves coverage but reduces lexical diversity by assigning borderline papers to nearest topics.

Model	Topics	Outlier	Largest	Diversity	c-NPMI	c_v
MiniLM selected, pre-reduction	56	0.345	0.067	0.945	0.144	0.740
Selected BERTopic + reduction	56	0.000	0.082	0.518	0.177	0.733
LDA TF-IDF, $K = 30$	30	0.000	0.062	0.763	0.042	0.526

6 Results: Topic Trends

The largest net movement is LLM reasoning (+4.19 percentage points), followed by 3D vision/scene generation (+2.30), preference learning/alignment (+1.55), LLM safety/jailbreaks (+1.54), and efficient attention/KV cache (+1.23). Declines are strongest in RL/policy learning (-2.46), graph neural networks (-1.87), bandits/regret (-1.68), matrix algorithms (-1.66), and adversarial robustness (-1.63). These are changes in accepted-paper share, not acceptance rates.

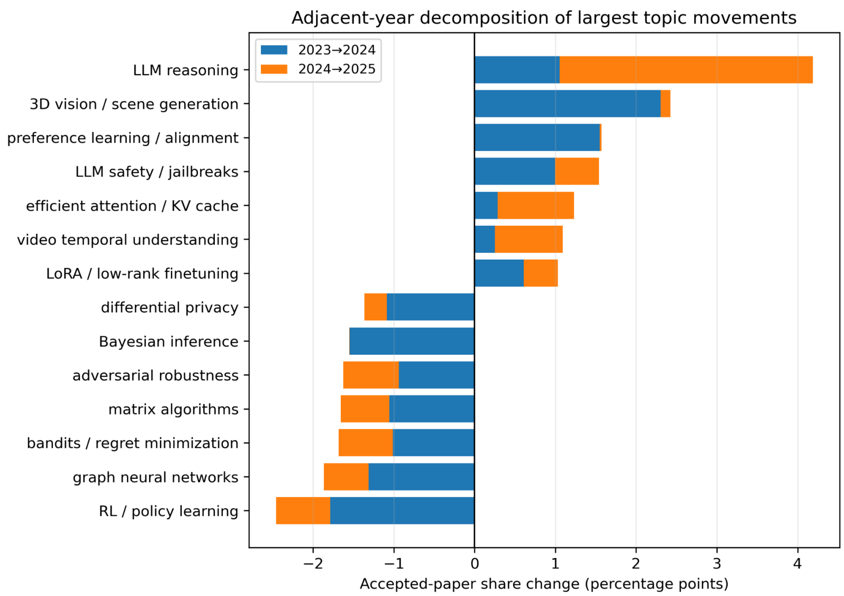


Figure 2: Adjacent-year decomposition of the largest topic movements. The bars separate 2023–2024 movement from 2024–2025 movement, showing year-specific shocks rather than only endpoint change.

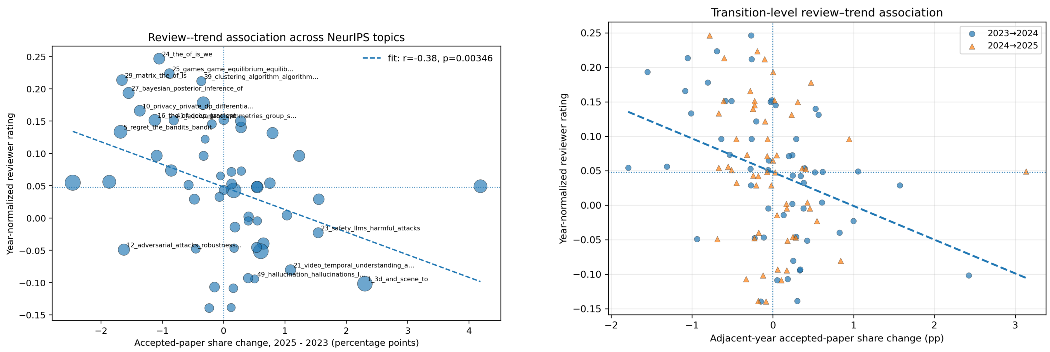
Table 3: Fastest rising and declining accepted-paper topics with adjacent-year decomposition. Shares are percentages of accepted papers in each year. Δ is the net endpoint change from 2023 to 2025.

Direction	Topic	Size	2023	2024	2025	$\Delta_{23\to24}$	$\Delta_{24\to25}$	Δ	z rating
Rising	LLM reasoning	486	1.77	2.83	5.96	+1.05	+3.13	+4.19	+0.049
Rising	3D vision / scene generation	882	5.28	7.71	7.59	+2.42	-0.12	+2.30	-0.102
Rising	Preference learning / alignment	200	0.44	2.01	1.99	+1.57	-0.02	+1.55	+0.029
Rising	LLM safety / jailbreaks	149	0.22	1.21	1.76	+1.00	+0.54	+1.54	-0.023
Rising	Efficient attention / KV cache	252	1.40	1.69	2.63	+0.29	+0.94	+1.23	+0.096
Declining	RL / policy learning	1029	9.82	8.03	7.36	-1.79	-0.67	-2.46	+0.055
Declining	Graph neural networks	507	5.25	3.94	3.39	-1.31	-0.55	-1.87	+0.056
Declining	Bandits / regret minimization	482	4.88	3.87	3.20	-1.01	-0.67	-1.68	+0.133
Declining	Matrix algorithms	201	2.64	1.59	0.98	-1.06	-0.60	-1.66	+0.214
Declining	Adversarial robustness	258	3.05	2.11	1.42	-0.94	-0.69	-1.63	-0.049

The decomposition shows that not all rising topics behave identically. LLM reasoning grows in both intervals and accelerates from 2024 to 2025. LLM safety and efficient attention also grow in both intervals. In contrast, 3D vision and preference learning jump from 2023 to 2024 and then plateau or slightly decline in 2025. The major declining topics are more consistent: RL/policy learning, graph neural networks, bandits, matrix algorithms, and adversarial robustness all decrease in both adjacent intervals.

7 Results: Review–Trend Association

Figure 3 shows the main review–trend evidence. In the endpoint analysis, topic net movement is negatively associated with year-normalized reviewer rating: Pearson $r = -0.384$ ($p = 0.00346$) and Spearman $\rho = -0.471$ ($p = 2.46 \times 10^{-4}$). Raw reviewer ratings produce a stronger association (Pearson $r = -0.707$, Spearman $\rho = -0.818$), but year-normalized ratings are our primary specification because they reduce year-specific calibration bias.



(a) Endpoint association.

(b) Transition-level association.

Figure 3: Review–trend association. The endpoint plot uses 2025–2023 net movement; the transition plot uses adjacent-year movements, 2023→2024 and 2024→2025.

Table 4: Endpoint and transition-level evidence for the review–trend association.

Analysis	Specification	Statistic / Coef.	p-value
Endpoint correlation	Pearson, z rating	-0.384	0.00346
Endpoint correlation	Spearman, z rating	-0.471	2.46e-04
Endpoint regression	Controlled OLS, HC3 SE	-5.903	2.53e-04
Endpoint bootstrap	Controlled OLS, 95% CI	[-8.73,-2.78]	–
Transition correlation	Pearson, z rating	-0.321	5.50e-04
Transition correlation	Spearman, z rating	-0.387	2.50e-05
Transition regression	Controlled, topic-clustered SE	-3.690	8.32e-05
Weighted transition	Topic-size weighted regression	-6.389	7.77e-08

Controlled endpoint regression. We estimate

$$\Delta_c = \beta_0 + \beta_1 rating_c + \beta_2 \log(n_c) + \beta_3 \hat{s}_{c,2023} + \beta_4 \pi_c + \epsilon_c.$$

This controls for topic size, initial 2023 share, and oral/spotlight concentration. The rating coefficient remains negative under both unweighted and topic-size-weighted models. Bootstrap resampling over topics gives a year-normalized coefficient mean of -5.73 with 95% CI [-8.73, -2.78].

Transition-level robustness. Because each NeurIPS year can introduce a different research shock, we also run a transition-level analysis with two observations per topic: 2023→2024 and 2024→2025. Let $g_{c,t\to t+1} = \hat{s}_{c,t+1} - \hat{s}_{c,t}$. We regress $g_{c,t\to t+1}$ on topic reviewer rating, topic size, prestige concentration, the topic's initial share in year t, and a transition fixed effect, with standard errors clustered by topic. The negative association remains under this adjacent-year formulation, so the review–trend association is not solely an artifact of using the 2025–2023 endpoint.

8 Qualitative Topic Validation

A topic model is useful only if its clusters are semantically meaningful. We selected representative papers for each major topic using topic membership, topic probability, decision tier, and reviewer score. Table 5 shows examples, and the full representative-paper table for all 56 topics is included in the submitted output package.

Table 5: Representative papers validating major topic labels.

Topic	Representative papers
LLM reasoning	SGLang: Efficient Execution of Structured Language Model Programs; Why think step by step? Reasoning emerges from the locality of experience
3D vision / scene generation	MeshFormer: High-Quality Mesh Generation with 3D-Guided Reconstruction Model; CAT3D: Create Anything in 3D with Multi-View Diffusion Models
Preference learning / alignment	Direct Preference Optimization: Your Language Model is Secretly a Reward Model; Decoding-Time Language Model Alignment with Multiple Objectives
LLM safety / jailbreaks	Robust Prompt Optimization for Defending Language Models Against Jailbreaking Attacks; Protecting Your LLMs with Information Bottleneck
Graph neural networks	Unified Graph Augmentations for Generalized Contrastive Learning on Graphs; Equivariant Machine Learning on Graphs with Nonlinear Spectral Filters
Differential privacy	User-Level Differential Privacy With Few Examples Per User; Oracle-Efficient Differentially Private Learning with Public Data

9 Robustness, Limitations, and Reproducibility

The main result is supported by several robustness checks. First, the negative association appears under both Pearson and Spearman correlation. Second, it remains negative after controlling for topic size, initial share, and prestige concentration. Third, bootstrap confidence intervals over topics exclude zero. Fourth, BERTopic improves substantially over LDA in coherence. Fifth, pairwise stability indicates that selected topics are not arbitrary one-off clusters: within-family NMI is around 0.73, while LDA-vs-BERTopic similarity is much lower. Sixth, adjacent-year transition regressions show that the result is not solely caused by endpoint definition. Finally, qualitative validation confirms that major topics correspond to recognizable research areas rather than opaque clusters.

This project should be interpreted as accepted-corpus data mining, not submitted-paper causal inference. The result says that topic growth in the accepted NeurIPS corpus is negatively associated with reviewer evaluation; it does not say that a topic has a higher or lower true acceptance probability. It also does not prove that reviewers are biased against emerging topics. Limitations include the incomplete rejected denominator, a three-year time horizon, approximate model-assisted topic labels, possible calibration differences in reviewer scores, and sensitivity to embedding choice. Because the unit of analysis in the main regression is topic rather than paper, the regression should be interpreted as topic-level association, not paper-level causal evidence.

The submitted code repository contains modular scripts for data preparation, embedding generation, BERTopic grid search, outlier reduction, LDA baseline, model evaluation, trend analysis, regression, representative-paper selection, and figure generation. The server run used CUDA-enabled PyTorch, cached transformer embeddings, and saved environment files (`environment.lock.yml` and `pip_freeze.txt`). A compact report-ready output package contains the tables, figures, logs, and evaluation summaries used in this write-up; a larger full-artifact archive stores model checkpoints, embeddings, processed data, and complete experiment outputs. Server-output checksums were recorded for the core CSV results.

10 Contributions and Conclusion

Table 6: Author Contributions and Responsibilities

Author	**Contributions & Responsibilities**
Nguyen Pham Tuan Anh	Implemented the OpenReview data pipeline and preprocessing,embedding generation, BERTopic grid search, outlier reduction, and model evaluation.
Tran Hung Dat	Trend metric computation and accepted-paper share analysis. Representative-paper selection and qualitative topic validation.
Tran Ho Chi Thanh	Implemented trend statistics, correlation/regression analysis, transition analysis, visualization, qualitative validation, and report writing.
All Members	Reviewed the final acceptance-rate correction and interpretation.

We upgraded the original milestone into a validated data-mining study of NeurIPS research trends. The final analysis removes invalid topic-level acceptance-rate claims and focuses on accepted-paper share, adjacent-year topic movement, reviewer score, and decision-tier concentration. BERTopic with outlier reduction produces more coherent topics than the LDA baseline, and representative papers validate the major topics. The main empirical finding is a review–trend misalignment: fast-growing accepted-paper topics tend to have lower year-normalized reviewer ratings, while several high-rated theoretical or classical learning areas decline in share. By decomposing 2023–2025 net movement into adjacent-year changes, we avoid treating endpoint movement as a long-term monotonic trend and provide a more defensible view of how NeurIPS research attention shifted over the observed window.

References

[1] Paper Copilot. NeurIPS Statistics. `https://papercopilot.com/statistics/neurips-statistics/`.

[2] OpenReview. Python API documentation. `https://docs.openreview.net/`.

[3] M. Grootendorst. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv:2203.05794*, 2022.

[4] BERTopic documentation. Outlier reduction. `https://maartengr.github.io/BERTopic/getting_started/outlier_reduction/outlier_reduction.html`.

[5] A. Singh, M. D'Arcy, A. Cohan, D. Downey, and S. Feldman. SciRepEval: A Multi-Format Benchmark for Scientific Document Representations. *arXiv:2211.13308*, 2022.

[6] D. Blei, A. Ng, and M. Jordan. Latent Dirichlet Allocation. *Journal of Machine Learning Research*, 2003.

[7] L. McInnes, J. Healy, and J. Melville. UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. *arXiv:1802.03426*, 2018.

[8] R. Campello, D. Moulavi, and J. Sander. Density-Based Clustering Based on Hierarchical Density Estimates. *PAKDD*, 2013.

[9] M. Röder, A. Both, and A. Hinneburg. Exploring the Space of Topic Coherence Measures. *WSDM*, 2015.